MUSEUM ARTIFACT EXPLORER — FINAL REPORT

Flask + SQLite Relational DBMS Hardening & Feature Expansion

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1. EXECUTIVE SUMMARY
The Museum Artifact Explorer was enhanced into a robust relational DBMS platform. Technical debt and unneeded files were removed, database schema integrity was enforced using CHECK, NOT NULL, and UNIQUE constraints, and M:N associations (curators, exhibitions, exhibition_artifacts), an AFTER DELETE audit trigger, multi-column indexes, and a denormalized SQL view (artifact_details) were added. The automated test suite was expanded to 16 unit tests, all passing cleanly.

2. DELIVERABLES MATRIX
• Phase 1 (Cleanup):
  - Added museum.db to .gitignore; removed from git cache.
  - Deleted package-lock.json, templates/details.html, static/style.css.
  - Cleaned README personal Windows paths and removed 'Antigravity IDE'.

• Phase 2 (Fixes):
  - Sanitized 'next' parameter in toggle_favorite against open redirects.
  - Made FLASK_DEBUG configurable from environment (default off); added SECRET_KEY warning.
  - Removed legacy destructive DROP TABLE migration logic in database.py.
  - Added NOT NULL on FKs, CHECK(discovery_year <= 2026), CHECK(length(trim(name)) > 0), UNIQUE(name, civilization_id); handled duplicate IntegrityError in add/edit routes.
  - Assigned distinct Unsplash URLs for Roman Denarius and Chola Gold Coin.

• Phase 3 (DBMS Add-ons):

DATABASE SCHEMA, TEST RESULTS & SECURITY

3. FINAL RELATIONAL SCHEMA OVERVIEW
• civilizations (id PK, name NOT NULL UNIQUE, region, period, description)
• categories (id PK, name NOT NULL UNIQUE, description)
• artifacts (id PK, name NOT NULL, description NOT NULL, period, discovery_location,
  discovery_year, material, image_url, civilization_id FK NOT NULL, category_id FK NOT NULL,
  created_at, CHECK(discovery_year <= 2026), CHECK(length(trim(name)) > 0),
  UNIQUE(name, civilization_id))
• favorites (id PK, artifact_id FK NOT NULL UNIQUE ON DELETE CASCADE, created_at)
• curators (id PK, name NOT NULL, email NOT NULL UNIQUE, specialization)
• exhibitions (id PK, title NOT NULL UNIQUE, start_date NOT NULL, end_date NOT NULL,
  curator_id FK ON DELETE SET NULL, CHECK(end_date >= start_date))
• exhibition_artifacts (exhibition_id FK, artifact_id FK, PRIMARY KEY (exhibition_id, artifact_id),
  both ON DELETE CASCADE)
• deleted_artifacts_log (id PK, artifact_id NOT NULL, name NOT NULL, deleted_at)
• View: artifact_details (artifacts JOIN civilizations JOIN categories)
• Trigger: trg_artifact_deleted (AFTER DELETE ON artifacts INSERT INTO deleted_artifacts_log)
• Indexes: idx_artifacts_name ON artifacts(name), idx_artifacts_civ_cat ON (civilization_id, category_id

4. UNITTEST EXECUTION SUMMARY
Ran 16 tests in 0.619s — Status: OK (All passed)
Tests covered: schema tables, idempotency, homepage, catalog search/filter, detail/404,
CRUD lifecycle, favorites cascade, civs/cats views, empty name validation, missing civ validation,
duplicate name rejection, open redirect sanitization, CHECK constraints, trigger logging,